

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

**Negative Exponent Law***Question Bank 3.04*

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*Show your work. Circle your final answer.*

**For questions 1 through 6, rewrite each expression as an equivalent expression with positive exponents. Do not evaluate.**

1.  $8^{-3}$

2.  $\frac{1}{(-4)^{-5}}$

3.  $(-2)^{-8}$

4.  $\frac{1}{10^{-2}}$

5.  $-5 \cdot 6^{-4}$

6.  $\frac{2^{-3}}{9^{-2}}$

**For questions 7 and 8, apply the negative exponent law, then the product or quotient of powers law, and evaluate.**

**7.** Evaluate each expression.

**a.**  $(-3)^2 \cdot (-3)^{-5}$

**b.**  $2^8 \cdot 2^{-4}$

**c.**  $9^9 \cdot 9^{-9}$

**8.** Evaluate each expression.

**a.**  $(-5)^{-6} \div (-5)^{-4}$

**b.**  $2^0 \div 2^{-3}$

**c.**  $3^{-7} \div 3^5$

**Opposite or reciprocal.**

9. Four expressions are shown:  $-4$  ,  $4^{-1}$  ,  $(-4)^{-1}$  ,  $-4^{-1}$  .
- a. Give the value of each one.

- b. Which of the four is the reciprocal of 4, and which is the opposite of 4?

**Working backwards.**

10. Find the value of  $n$  that makes each statement true.

a.  $2^n = \frac{1}{32}$

b.  $10^n = 0.001$

c.  $7^3 \cdot 7^n = 7^{-1}$

11. Which is the value of  $4^{-3}$  ?

- ☐ A.  $\frac{1}{64}$
- ☐ B.  $-64$
- ☐ C.  $-12$
- ☐ D.  $\frac{1}{12}$

12. Select ALL of the expressions whose value is  $\frac{1}{64}$  .

- ☐ A.  $2^{-6}$
- ☐ B.  $4^{-3}$
- ☐ C.  $8^{-2}$
- ☐ D.  $\left(\frac{1}{2}\right)^6$
- ☐ E.  $-2^6$
- ☐ F.  $2^{-8}$